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Research Article

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An Effect of Antarctic Ice Melt Towards Seawater Density, Coastal Stability and Seismic Activity

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Abstract

In the recent century, the issue of raising sea level is becoming higher and higher prominent due to the world's rising urbanisation. Combinations of structural, anthropogenic, and climate-related variables contribute to sea-level rise. This publication reviews and analyses recent research on these elements' contributions to raising sea level. The findings indicate that melts glaciers and sheets of ice has primarily caused raising sea level, and we expect this trend to persist in the upcoming decades. As the problem gets worse, geological disasters linked to rising sea levels are getting greater attention. Research suggests that the ocean's ice melting will bring about 2.6% more water into the system than the sea initially displaced. The ice that appears to be melting due to global-warming will result in an increase in the ocean level. All the remaining floating shelf ice and sea ice will melt. The water level would rise by around 5 cm if it melts. When grounded snow slides into the ocean, it causes a two-part indicates rise in level of water, with some of the rise occurring later. When the ice is floating, the initial portion is equivalent to the amount of sea water that has been displaced. The following segment is equivalent. As it melts, it contributes 2.6% of the first. These outcomes of the difference in the equivalent weight quantity between fresh and salt water. This aspect of raising sea level, which can be seen as a type of sea level rise caused by halostronicin relation to the melted (even before it melts) and displaced salt water collectively, appears to have gone unacknowledged in the literature up to this point. Variations in salinity are recognised for impact level of the sea, yet all current analyzes overlook onvolume of estimated shift. We place additional review and emphasis on Antarctic Ice Melt, raising Sea-level and Ocean Density Changes, Geological Stability, Antarctic Continental Shelf Fracturing, and Seismic Activity to gain a deeper comprehension the impact of raising sea level. We proposed the Advanced Archimedes Principle Based on Enhanced Conditional Variational Autoencoder (AAP-ECVAE) to address the threats posed by rising sea levels disasters. We ultimately subject the coastal urban system to a resilience-based quantitative disaster assessment concept, which evaluates its resilience to geological disasters resulting from sea-level rise. When compared to the RF, GP, and NN models, the suggested model AAP-ECVAE shows the best accuracy, precision, and recall.

Keywords:raising sea level, Antarctic ice melt, Geological Stability, Seismic activity, AAP-ECVAE.

1. Introduction

One of the most important factors influencing how humans will develop in the future is the rising in sea level. Around 45% of the The majority of people on Earth reside in island nations and coastal regions could be submerged due to raising sea level [1]. Raising sea level has the potential not just submerge areas along the coast but also cause seismic disasters that typically outcome of property lack and fatalities [2]. As a result, some residents of coastal zones have been forced to relocate from areas where seawater poses a threat. Raising Sea level is one of the most significant effects of climate change. The high projected rates of sea level rise in the future have caught the attention of people worldwide. Particularly small island nations and low-lying locations fear that coastal erosion and floods may shrink their land area and, in the worst case, drive a sizable portion of their population to migrate overseas. Therefore, as the implications of climate Changes starts to become more noticeable, this topic has garnered increased attention on a global scale[3]. The sea level varies over time and space due to natural phenomena like tides and waves. Mean sea level at a given site is the height of the sea surface averaged over a period of time, say a month or a year, long enough to minimise tide and wave oscillations. There is also a regional distribution of the global mean sea level. The average sea level throughout all of the world's seas is known as the global mean sea level, or GMSL. Because regional and local scale factors influence the global mean change, local mean sea level changes typically diverge from GMSL trends Sea-level change (rise), as used in this work, refers to the overall change (raise) in mean sea level, taking into account both local and global influences [4].

The steady increase in the planet's average surface temperature, commonly known as "enhanced global warming," is primarily about human endeavours such as the removal of forests and the burning of fossil fuels. The accelerated rate of glacier melting, particularly in Antarctica, is among the most serious effects of global warming. There has been an unprecedented volume of ice loss of the ice sheet in Antarctic, which houses around 60% of the freshwater on Earth. In addition to contributing to the rise in sea levels globally, this melting is having broader effects on the tectonic and geophysical regimes of the planet Earth[5]. The following figure 1 can be used to summarise the implications of global warming:

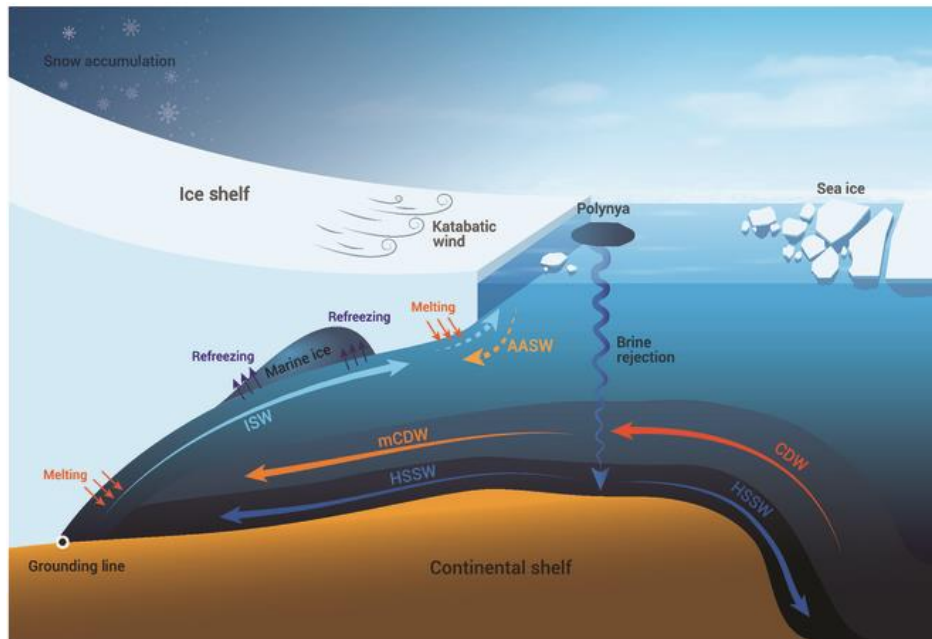


Figure 1: The key impact of global warming

Earth's largest freshwater reservoir is the Antarctic ice sheet, the majority of which is located in East Antarctica. Global sea levels might rise by 60 meters if it all melts. According to current projections, sea levels will increase by one meter by 2100 and by more than fifteen meters by 2500. Therefore, it is critical for scientists to comprehend the Antarctic continental ice melting process in order to make more precise sea level predictions. The majority of research on the interaction between ocean and ice has been done on West Antarctic ice shelves[6]. Because it was previously believed that most of the East Antarctica's ice shelves had cold water chambers keeping them from melting, these shelves have gotten far less attention. Between January and February of 2017, Daisuke Hirano and associates conducted a research excursion that yielded data on water temperature, salinity, and oxygen levels from 31 locations in the region. To comprehend ocean circulation beneath the Shirase Glacier Tongue at the glacier's inland base, they integrated this knowledge with information on the currents and wind in the area, ice radar observations, and computer modelling. According to the scientists' findings, deep, warm water is moving inward and towards the base of the Shirase Glacier Tongue, which is causing the melting[7]. Warm water rises at the base of the tongue after travelling in a deep ocean trough, melting and warming the ice. After that, the warm waters carrying the melting ice flow forth and combine with the meltwater from the glaciers.

The researchers discovered that while this melting happens all year round, it is influenced by seasonally variable easterly alongshore breezes. The summertime decrease in winds causes an increase in the inflow of warm, deep water, which accelerates the melting process. According to recent studies on climate change, post-glacial rebound causes the weight of the Antarctic ice sheet to lessen as it melts, hence decreasing the earth's crust. The base of the glacier begins to float when the bulk of the ice gradually diminishes, which might alter the tectonic plate's functions and the availability of mantle material. Moreover, the seawater's dilution due to the melting glaciers has a negative impact on its salinity and density, which alters the thermohaline circulation pattern. This circulation is an important part of the worldwide climate system that supports weather patterns and ocean currents, among other things. The distribution of stress on tectonic plates within the seas is also influenced by changes in ocean circulation and density, especially in subduction zones, where two plates collide.

Overview of Seismic Activities

The shifting and movements of the lithosphere, or the hard outer layer of the Earth divided into several major and smaller plates, are referred to as tectonic activity. These plates lie atop the semi-molten asthenosphere, and over a long geological period, shape in their movements with the features of the surface of earth [8] In addition to causing earthquakes and volcanic eruptions, geologic plates are crucial in the development of mountains, oceanic trenches, and volcanic islands. These plates are moving because of internal mantle forces like convection currents brought on by heat from the Earth's core. The three primary categories of these movements are transform, convergent, and divergent. Therefore, it is crucial for both the development of geological formations and the management of seismological hazards on Earth as a whole. Tension that has progressively built up along the Earth's crust releases itself unexpectedly during earthquakes; plate boundaries are often the site of this phenomena[9]. A zone of subduction is where both plates meet driven beneath one another and are therefore highly seismically active. For example, because of its dense concentration of convergent and transform boundaries, the Pacific Ring of Fire is a prominent area of the Pacific Coast that hosts numerous earthquakes and volcanic fields. Japan is highly vulnerable to earthquakes because it is situated at the meeting point of multiple plates with tectonic, including the Plate with pacific, the Sea Plate with Philippine, and the Plate with Eurasian [10]. Information on the composition, potential modifications, and potential influences on tectonic forces For example, managing and preventing disasters depends on climate change.

Objectives of the Research

- The aim of this research is to ascertain whether a relationship exists between tectonic activity and Antarctic melting in relation to seismic activity in the vicinity of Japan. The following theories serve as the study's compass.
- Water from the Antarctic ice shelf dilutes seawater, changing global thermohaline circulation and increasing tectonic forces in subducting regions, including those near Japan. These effects of melting oceans also affect surface pressure and the mantle's convection pattern, which in turn affects the global pattern of tectonic stress.
- The pressure of the water column on the associated coastal and seafloor areas may cause seismic activity at the associated tectonic fault zones near Japan to activate or perhaps intensify when sea levels rise as a result of Antarctic ice melting.

2. Related Works

- Topographic information and subsurface oceanic, glacial, and hydrological processes constrain the accuracy of numerical models used to study the evolution of the Antarctic system. It is important to make direct remarks in difficult places, but it is not appropriate to cross disciplinary lines [11].
- The paper offers mitigation strategies, presents a quantitative disaster assessment approach for coastal urban systems, and looks at how raising sea-level affects disasters in geological such as storm surges, underwater mass failure, seismic activity, seawater intrusion, and destruction of coastal wetlands[12].
- We propose that polar ice sheets appear to have changed as a result of the low latitude climate anomaly that was carried by the late Holocene atmosphere up to the high latitude [13].
- In this synopsis, the causes, effects, and implications of raising sea level are discussed from both a global and Japanese viewpoint, along with ways to mitigate its negative effects [14].
- This paper recommends building a multiple source monitoring system for incursion of seawater and provides a scientific foundation for using groundwater for regional productivity and living [15].
- According to the study, increased seawater content rather than temperature is the primary factor controlling fracture detainment in Antarctic suture zones, which in turn controls mass balance [16].
- In order to make clear the implications of raising sea level for human's society, this study evaluates the literature on the topic. It covers its causes, future projections, affects, and

consequences from both a global and Japanese viewpoint. It also discusses remedies to adverse impacts [17].

3. Methodology

Thus, the research is based on quantitative analysis and model simulations presented in this chapter, which integrate unique data and statistical techniques to thoroughly examine the relationship between Antarctic ice melt Towards Seawater Density, Coastal Stability and Seismic Activity in Japan

3.1 Data Collection

The upper-atmosphere sounding database at the University of Wyoming, located at (<https://weather.uwyo.edu/upperair/sounding.html>), provides wind speed and SWEAT index data for all 365 days at 2000 meters above sea level. The data is for diurnal and re-analyzed scales. The Meteorological Agency of Japan (<https://www.jma.go.jp/jma/en/Activities/upper/upper.html>) has information on the dispersion of Japan's radiosonde network, including the locations of the stations at Kagoshima and Hachijojima. Primary upper-atmosphere data for six seismic events are gathered from the Hachijojima station in each of these case studies, with additional data being added from the Kagoshima station to round out the dataset. Ultimately, the daily precipitation time series for Hachijojima (32–33° North, 138–141° East) and Kagoshima (30–31° North, 131–133° East), which had already been processed for the Tropical Rainfall Measurement Mission (TRMM) project, were obtained from the Asia Pacific Data Research Centre (APDRC) at (<https://apdrc.soest.hawaii.edu/las/getUI.do>). Using the 4th version of the Online Geospatial Interactive Visualisation and Analysis Infrastructure (GIOVANNI) program (<https://giovanni.gsfc.nasa.gov/giovanni>), which is followed by the NASA Goddard Earth Sciences Data and Information Services Centre, a visualisation of the TRMM precipitation data was produced as maps with a time average. The composite anomaly of the geopotential height at 500 hPa was mapped using the NOAA/Physical Sciences Laboratory daily reanalysis database, which assisted in validating specific storm centres based on the standard climatology period utilising (<https://www.esrl.noaa.gov/psd/data/composites/day>).

Additionally, information from the Global Volcanism Program at NMNH in New York was used to distinguish orange seismic occurrences and those brought on by volcanic outbursts in the research area

: https://volcano.si.edu/search_eruption.cfm. In 2017, three recorded volcanic eruptions were identified in this region: Aira (code: 28208), Kirishimayama (code: 282090), and Nishinoshima (code: 284096). The

Volcanological Society of Japan and the Japan Meteorological Agency provided data from the 4th edition of the worldwide Catalogue of Active Volcanoes in Japan, which was used to map the geographic distribution of volcanoes and calderas in Japan and its surroundings (https://www.data.jma.go.jp/svd/vois/data/tokyo/STOCK/souran_eng/menu.htm).

3.2 Preparing Data

Radon emission and electromagnetic fluctuations

Using GAMMA software, the ALOS-2/PALSAR-2 data used in this investigation was processed. The technique of two-pass differential interferometry was employed. The SAR images of all three tracks have an azimuth resolution of 2.12, 1.87, and 1.85 meters for the DL, AR, and DR tracks, respectively, and a range resolution of 1.43 meters. We used multi-look factors of 15×17 for azimuth and range, respectively, to maintain pixel spacing of about $31 \text{ m} \times 31 \text{ m}$ in the ground range and azimuth directions, ensuring excellent interferometric coherence. Using the 31 m resolution Global Digital Surface Model ALOS World 3D-30m AW3D30 from JAXA, which is available at <https://www.eorc.jaxa.jp/ALOS/en/aw3d30/data/index.htm>, the topographic phase contribution was modelled and eliminated. Next, we used the MCF feature of the GAMMA program to unravel the interferogram. Furthermore, using high coherence far-field measurements as a basis, a bilinear approach was developed for modelling and adjusting orbital error contributions. As a consequence, coseismic surface deformation fields were precisely retrieved in the LOS directions.

3.3 Data Analysis

The determination of CCF with diurnal time-lag correlations is the first step in the statistical data analysis in the current study. This would provide an estimate of the time lags between atmospheric predictors and seismic indicators, which can range from several days to the point at which unusual impacts or stress build-up in the environment become apparent. The current study examines three atmospheric pre-earthquake variables: the rate of precipitation, wind velocity at 2000 meters above sea level, and the index in SWEAT. The SWEAT index is arguably the most important variable for weather prediction out of all the ones mentioned. By utilising kinetic thermodynamics and radiosonde data, it is able to identify air disturbances and severe weather. The dimensionless SWEAT index can be represented as follows, as the authors have shown:

$$SWEAT = 12T_{d850} + 20(TT - 49) + 2f_{850} + f_{500} + 125(S + 0.2)$$

where (T_{d850}) is the point temperature at 850 hPa in °C, while (f_{850}) and (f_{500}) are the wind speeds at 850 hPa and 500 hPa, respectively, in knots. (TT) is the index for air temperature between 850 hPa and 500 hPa in °C. $\angle(S)$, the shear or slope, between 850 hPa and 500 hPa, depicts wind direction, from 0 to 360°. SWEAT values are categorized into three weather severity categories: low (< 200), high (200–300), and extreme (> 300).

The research culminates in the application of a multidisciplinary metric known as the Climate with SeismoIndex (SCI), which is derived from variations in earthquake precursors both before and after atmospheric event forecasters:

$$SCI = (Fre_i \times Mean_i \times Maxi) / (Fre_j \times Mean_j \times Max_j)$$

where SCI is the index, (Fre_i) and (Fre_j) are the frequency of seismic activity occurring after and before the selected storm events, respectively. $Mean_i$ and $Mean_j$ are the mean magnitudes of earthquakes occurring after and before the storm events, and $MaxA$ and $MaxB$ are the maximum magnitudes of both during and prior to the storm events, earthquakes. Among these, the storm occurrences in this study were selected based on proven lag-day intervals inferred using the CCF approach, which were set at 8-9 days or 14-16 days ahead of the major shocks. SCI values greater than 2.00, and especially those greater than 3.00, suggest that there will be a reliable connection between the storm events and the ensuing seismic activity.

Clusters of lesser earthquakes, also known as swarm clusters, usually occur after major earthquakes ($M > 6$). A swarm cluster is a collection of earthquakes that occur in the time and space that follow the main shock. The frequency and size of these minor seismic activity within the 30 days before and after the chosen storm episodes, which occur 1-2 weeks before the main shocks, are taken into account by the Seismo-Climatic Index (SCI) in the study. These were utilised in the article because pre-earthquake air ionisation events resulting from anomalies created with atmospheric thunderstorms are frequently responsible for greatly elevated seismic activity, including both the primary shocks and following swarm clusters.

3.4 Advanced Archimedes Principle Based on Enhanced Conditional Variational Autoencoder (AAP-ECVAE)

3.4.1 Applying Advanced Archimedes' principle

The density of ocean ice, or sea water that has frozen, is around 2000 kg m^{-3} when it melts and approximately 918 kg m^{-3} while frozen. It is noteworthy that the density of the ice has no bearing on the argument as long as it remains afloat. The occasional presence of brine inclusions in the ice has relatively little impact. The ice shelves are incredibly low in salinity. Sea ice is created when seawater freezes, and while it is mostly cleansed through "brine rejection," it still contains some salt. Particularly in first-year sea ice, some brine inclusions are still present. For the sole factor that matters is the meltwater's density in our investigation, air bubbles that are either suspended in shelf or sea ice have no bearing on the results. Since shelf ice makes up roughly 10 tms the amount of ocean ice, the brine inclusions do have a slight impact on the meltwater density, but this effect will not be significant for our findings. However, result will take into account the impact of the brine inclusions by dividing the sea bulk ice by a salinity-based factor, treating it as though it were composed of both clean sea water and ice (the latter of which has neutral buoyancy). The adjustment won't be noticeable. Ocean surface water has a density of roughly $\rho_0 = 1026 \text{ kg m}^{-3}$. It is considered that a giant iceberg or chunk of floating sea ice floating with volume V_i above the mean waterline and V_j below it.

$$V = V_i + V_j \quad (1)$$

The ice's weight (W) is

$$\text{Weight } (W) = \rho_i g V = \rho_i g a (V_i + V_j), \quad (2)$$

where ga is the gravitational quickening By Advanced Archimedes' Principle,

$$W = \rho_0 g a V_j. \quad (3)$$

The amount of meltwater that remains after melting will be

$$V_{\text{melt}} = W / (\rho_W g a) \approx W / (2000 g) \quad (4)$$

Combining both Equation (3) and (4) we find

$$V_{\text{melt}} = V_j \rho_0 / \rho_W \approx 1.026 V_j \quad (5)$$

As a result, the water level rises since the quantity of melted water is 2.7% greater than the amount of ocean water that was displaced. There is a very slight volume contraction caused by mixing when clean meltwater combines with the salt water, of about 0.1 percent or less, but it is insignificant in comparison to the rise shown here. The recent volume rise is in line with Archimedes' Principle, which states that weight, not volume, is the subject of the rule. The weight of the underlying liquid is displaceable by the

ice, but it does not move the same amount of material as melted water. Our analysis of rising sea levels is only accurate to about 11% due to a number of factors, including changing physical conditions, the combination of data from multiple sources (spacecraft and submarines), measurement errors in these data, the presence of poorly determined impurities on floating ice, the variability of salinity in the sea water that provides flotation, and our failure to take into account the contribution of melting ice to thermosteric changes. Rather than using explicit error bars in our volumetric computations, we only carry a certain number of significant digits.

3.4.2 Enhanced Conditional variational autoencoder (ECVAE)

An encoder and a decoder are coupled in the popular AAP-ECVAE neural network architecture, where the intermediary layer among them represents a lower-dimensional latent space. Constructing generative models, such as a zero-shot learning generative model and a molecular generative model, is an attractive use of the ECVAE. In our attempt to develop a generative strategy for the simulation of sea level change, we take into account an AAP-ECVAE in addition to the feed-forward NN. This is because, by selecting a sample from the relevant latent space, one can produce rather than just one prediction for a given input, this method produces samples of predictions; in fact, it serves as a model for generative artificial intelligence (AI) methods like model of diffusion. The autoencoder (AE) and the variational autoencoder (VAE) are the sources of the ECVAE. Typically, an artificial intelligence (AE) is trained to replicate the initial input, which enables it to acquire a reduced-dimension latent representation that encompasses a significant portion of the higher-dimensional input data. Despite having a similar design to the CAE, the ECVAE makes the assumption that a certain distribution, often a multivariate normal with a mean of 0 and an identical covariance matrix, exists in the reduced-dimensional latent space.

The primary benefit of the ECVAE over the VAE is its ability to function as a generative model through sample extraction from the latent, intermediate layer distribution. Though they are similar in nature, the ECVAE has one more feature: it can tell the models to anticipate the outcome based on information on conditional. Combining conditional information with a random draw from the latent space (in our example, a change in the ice-thickness vector essentially yields a forecast for a new input. To create a prediction, the decoder processes this concatenated vector, which incorporates conditional and latent information. Using the whole ice-thickness vector varies as conditional data, our version of the ECVAE for the sea-level change simulator concatenates on latents sketch and then runs the vector concatenated via the decoder to provide a forecast. The encoder is made up of $L = 4$ hidden layers and 203 set of parameters $(\mu_i, \log \sigma^2_i)$, where $i = 1, \dots, 300$. The neurones in the layers are $(d1, d2, d3, d4) = (3200, 1600, 800, 400)$. These values match those of the spread of the posterior intermediate layer $P(Z|X)$, which is thought to have a diagonal covariance matrix $\text{diag}(\sigma^2_1, \dots, \sigma^2_{200})$ and a mean vector $\mu =$

($\mu_1, \dots, \mu_{200}$) that follow a multivariate normal distribution. Subsequently, the converted samples from the preceding distribution $P(Z)$ for the ice-sheet thickness vector (i.e., the conditional information) and the concatenated vector of the intermediate layer, are fed into the decoder. Re-parametrization $Z_i | X = \mu_i + \sigma_i Z_i$ is used to accomplish the transformation, assume $P(Z)$ has an identical covariance matrix and a multivariate normal distribution with a 0 mean. The decoder has $L = 5$ hidden layers and the encoder has same number of neurones. The estimated ocean level change is provided at twenty eightsites of long-term tide gauges. This is a variation of the standard AE approach since the CVAE is forecasting sea-level change values rather than recreating its own input. Two terms make up the training loss function: KL-divergence as first, or regularisation penalty, between the previous distribution $P(Z)$ and the variational approximation of the posterior latent distribution $P(Z|X)$. The second term is the mean squared error for the decoder's predictions. In our implementation, the prediction generation process takes 507 samples from the latent distribution $P(Z)$, and the mean of these 500 predictions is used to get the final predicted value. Python version 3.9.17 and PyTorch version 2.0.1 are used to implement the AAP-ECVAE.

$$\hat{y}_{ij}^{test} \pm t_{1-\alpha/2, n-2} \times \sqrt{S_i^{2, val} \times \left(1 + \frac{1}{n} + \frac{(\hat{y}_{ij}^{test} - \bar{y}_i^{val})^2}{\sum_j (\hat{y}_{ij}^{val} - \bar{y}_i^{val})^2} \right)}, \quad (6)$$

where $\hat{y}_{ij}^{test}$ represents the ML model's prediction for the i th location and j th test sample, where $i = 1, \dots, 28$ and $j = 1, \dots, 200$.

$$\hat{y}_{ij}^{test} = \hat{\beta}_{i,0}^{val} + \hat{\beta}_{i,1}^{val} \times \hat{y}_{ij}^{test}, \quad (7)$$

where $\hat{\beta}_{i,0}^{val}$ and $\hat{\beta}_{i,1}^{val}$ are obtained by using the validation set to fit a straightforward linear regression. Additionally, $S_i^{2, val}$, the i -th position, yields a variance estimate that is calculated by

$$S_i^{2, val} = \frac{1}{n-2} \sum_j (\hat{y}_{ij}^{val} - \bar{y}_i^{val})^2, \quad (8)$$

Predictive Evaluation of Japan's Earthquake Conditions

Japan lies in the most seismically active region of the world, located on the ring of fire in the pacific, the meeting point of 5 main plates with tectonic. Japan frequently experiences strong earthquakes, hence it is Developing efficient models to predict these types of natural disasters is crucial. Predicting the conditions of earthquakes is a difficult challenge that involves understanding the mechanics underlying the

occurrence of earthquakes, identifying precursors, and analysing the historical data. This section goes into more detail about these elements, particularly in relation to Japan and how they might be applied to enhance earthquake prognostication [18].

5. Results and discussion

The comparison of prediction interval lengths, empirical prediction interval coverage rate, and accuracy calculated on the test hold-out set comprises our primary findings. As a measure of accuracy, we compute the Advanced Archimedes principle coefficient of determination between the actual Values of sea level change simulations and the calibrated Machine Learning model prediction. Figure 2 clearly illustrates how the Random Forest (RF) model performs worse than the AAP-ECVAE, Gaussian Process (GP), and Neural Network (NN) models in terms of prediction accuracy. For every model, the average R^2 values are 0.844 (RF), 0.971 (GP), 0.989 (AAP-ECVAE), and 0.972 (NN). This indicates that the widely-used GP-based emulators may have competition from the AAP-ECVAE and NN. The length of prediction intervals and the range of $(2-\alpha)\%$ coverage on empirical rates for $\alpha = 0.01-0.15$, $0.05-0.15$, and 0.15 significance levels. Figure 4 shows that, upon calibration, Compared to other ML approaches, the empirical coverage rates for the GP model underperform the nominal interest rate considerably. Remarkably, even following the calibration, the Random forest still shows a little actual reporting variation from the nominal rates. This results from the RF model's less precise prediction performance. Although the goal of calibration is to improve point estimates and prediction interval accuracy, its efficacy may be weakened if the underlying model has poor prediction performance. Overall, in terms of length, coverage, and accuracy, the NN's prediction intervals and AAP-ECVAE are rather comparable. The AAP-ECVAE seems to be acting nearly indistinguishably from the NN; we think this is because both models share the same decoder architecture and the conditional ice-thickness input, which provides a greater amount of information than the latent variable used in the prediction. When all three of the analysed qualities are taken into consideration, the NN and AAP-ECVAE models appear to perform the best overall. We use the mean thickness of ice in several regions of Antarctica as inputs to evaluate the prediction performance of machine learning algorithms. With the exception of utilising each of the 18's mean thickness values zones as an input, the model structure is the same as that which is explained in Section 3. Furthermore, we see that all models' empirical coverage rates differ from their nominal counterparts. At a significance level of $\alpha = 0.05$ For example, for $\alpha = 0.06$, the empirical coverage rates are 0.913 (NN), 0.903 (AAP-ECVAE), 0.934 (GP), and 0.928 (RF), respectively, while the average prediction intervals are substantially broader (e.g., 0.171 (NN), 0.179 (AAP-ECVAE), 0.180 (GP), and 0.159 (RF), respectively. This implies that in order to maintain the information included in the input data, more

sophisticated dimension-reduction algorithms had to be used.

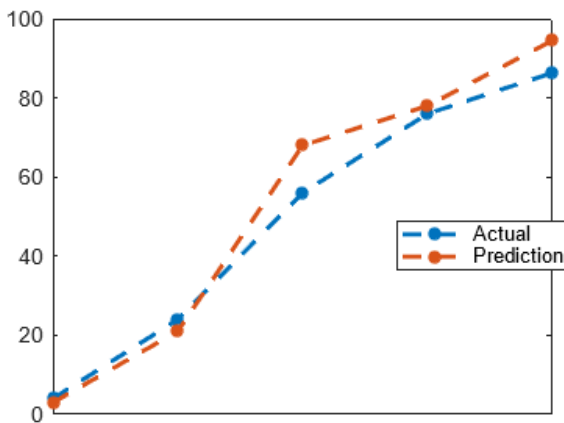


Figure 2: The Prediction accuracy of the Suggested method

Predictive Analysis Using Atmospheric Storms and Antarctic Ice Melt

Between 2000 and 2020, satellite monitoring revealed an unprecedented level of Antarctic ice melt. Japan saw historic seismic events during this time, including the 2011 Tohoku earthquake. Third Stage: During this phase, intervals of atmospheric storm activity were distinguished within the previously mentioned time limits. By examining the arrangement in the time series, several storm occurrences were discovered again prior to major earthquakes, as seen in the spatial visualisation of composites' reanalysis maps for 500 hPa geopotential height anomaly. For instance, the storm event that took place on April 25–26, 2017 (Figure 3a) happened roughly 13–14 days prior to the May 9, 2017, earthquake. In a similar vein, a storm event that occurred on July 18–19, 2017, was recorded 7–8 days prior to the July 26, 2017, earthquake (Figure 3b). Hence, all storm occurrences are discovered three, four, seven, or fourteen days before significant earthquakes. It is important to note that prior to each of these earthquakes, low-pressure anomalies with amplitudes ranging from -50 m to -250 m were clearly visible across the Japan region on the synoptic-scale climatological maps. The processes of blocking or perturbations to the anomalous tropospheric Rossby waves at mid-latitudes can explain these low-pressure episodes. The contact between the lithosphere and atmosphere after extensive air ionisation at the ground-to-air interface prior to big earthquakes may be the cause of the perturbations detected over seismic zones.

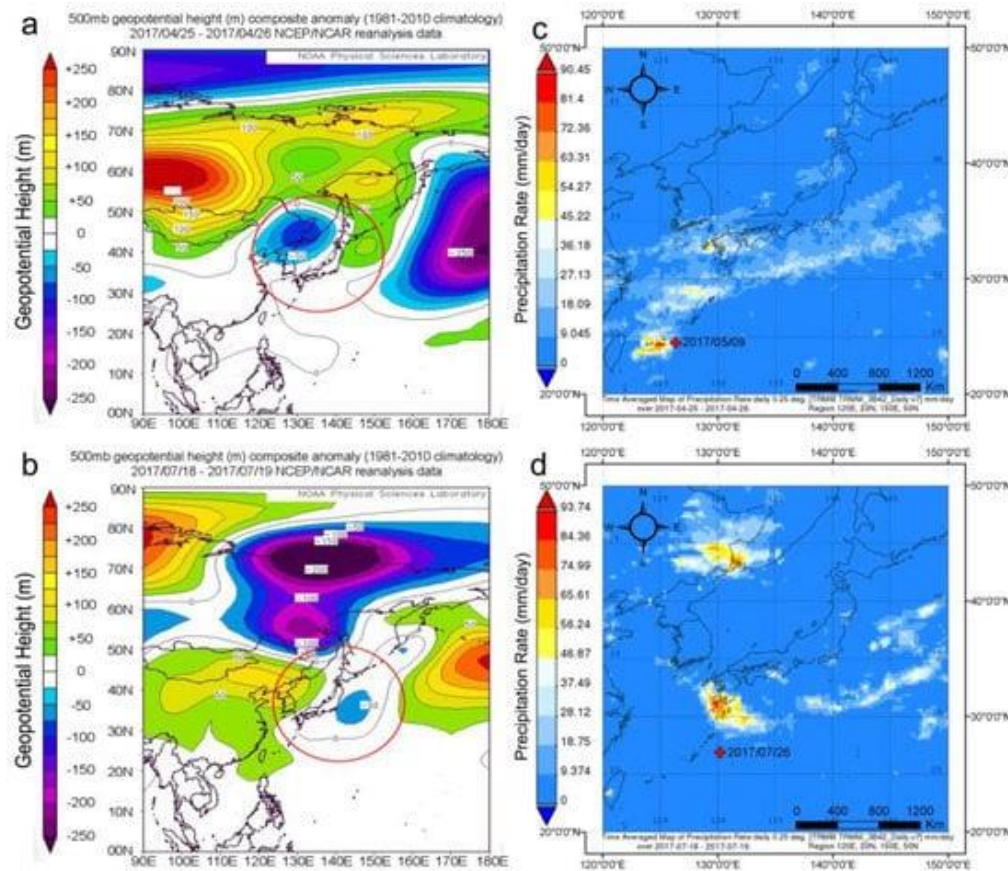


Figure 3: Reanalysed remote sensing maps visualised spatially

Temporal avg precipitation maps were produced for the chosen storm episodes using high-resolution TRMM precipitation data. Figure 3c displays a temporally averaged plot of the storm's daily precipitation rate event on 25–26 April, 2017, which occurred 14–15 days prior to an earthquake on 9th May, 2017. Additionally, for the storm event that occurred on July 18–19, 2017, or 8–9 days before the earthquake that occurred on 26th July 2017, maps showing the daily precipitation rate averaged throughout time have been created (Figure 3d). Within around 400 km from the epicentres of the earthquakes, distinct hotspots of precipitation have been recorded, with an estimated amount of 80 to 140 mm each day. This discovery would suggest a relationship between pre-earthquake signals and the sites of upcoming earthquakes. These findings also agree with the two characteristics that laboratory tests have identified as characteristic of positive hole charge carriers: i) In fact, similar to ultra-fast optical electric field probing, the positive hole charge carriers could escape the volume of strained rock and swiftly move over lengths of up to 100 km into less stressed or unstressed rocks. (2) Among nearby O²⁻ ions, the electronic wave function linked to positive holes has a tendency to delocalise on an atomic scale, altering the bonding

characteristics and, consequently, the physical characteristics of the rocks, such as their softness or plasticity under directional force. Predictive models were created by utilising this association to assess the potential seismic activity that could arise in the future due to ongoing glacier melt. It is concluded that with further decrease or loss of Antarctic ice, redistribution of stress may increase the likelihood of significant seismic events in the Nankai Trough and Sagami Trough regions. Because of this, the research provides a means of measuring earthquake recurrence intervals and potential magnitudes through the analysis of tectonic stress variations occurring from glacier melting.

Electromagnetic Variations and Radon Emissions as Predictive Tools

The monthly fluctuations of the electronic and ionic data that were observed at night over the Chinese mainland during the January 8, 2022, Menyuan MS 6.9 earthquake are shown in Figures 4 and 5. When the hourly Kp index reached 4 or the daily Kp index exceeded 3, certain data from December 20–21, December 27 and January 1, 3, and 9 are not included in the weekly processing. Between December 14 and December 20, 2021, ion density in northern China displayed notable anomalies, akin to those depicted in Figure 4a. The next week, ion density changes are depicted in Figure 4b while electron density fluctuations are shown in Figure 5b. When electron density was taken into account, no notable changes were seen between December 14 and 20. Since the background values for both parameters are rather high in northeast China, where these oscillations were recorded, the validity of these anomalies in the processed data is called into question. Moreover, no notable fluctuations have been observed over the Chinese mainland between January 4, 2022, and January 10, 2022—the date of the Menyuan MS 6.9 earthquake.

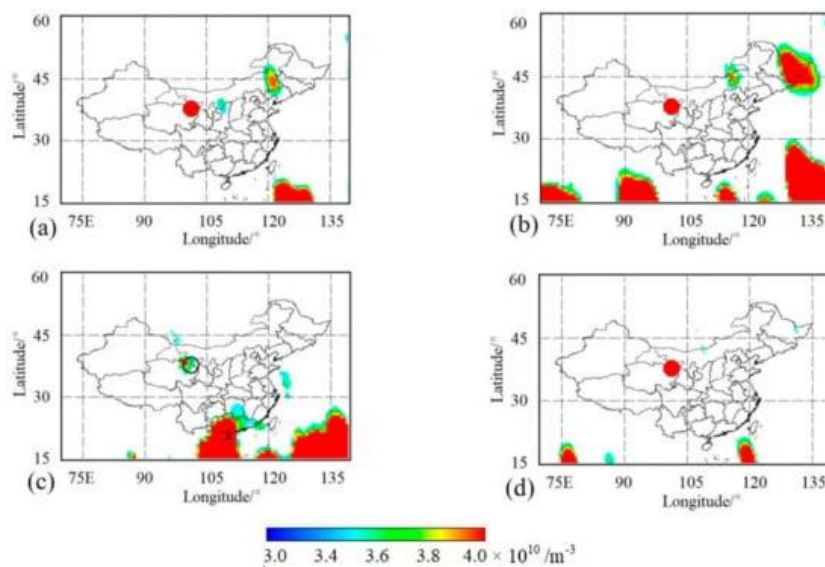


Figure 4: O+ density variations from 2021 to 2022

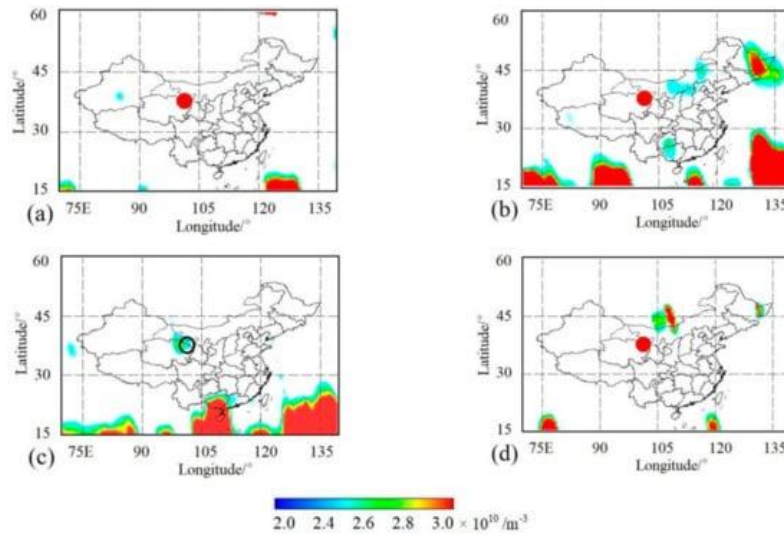


Figure 5 shows how the electron density changed from 2021 to 2022.

We were interested in the ionospheric fluctuations that occurred between December 28, 2021, and January 3, 2022. The processing of data began on January 5, 2022. The conditions for electron density and ion density are shown in Figures 4b and 5b, respectively. Mastery. Initially, in order to confirm that these were legitimate anomalies, we looked at satellite orbits that traverse the impacted area by pinpointing the precise orbit on which the abnormality in the signal was found.

Table 1:Accuracy, Precision and Recall of Existing and suggested methods

Methods	Accuracy (%)	Precision (%)	Recall (%)
RF [19]	86.21	84.85	86.02
NN [20]	90.43	89.72	90.12
GP[21]	79.84	78.35	79.21
AAP- ECVAE[Proposed]	96.28	94.63	96.25

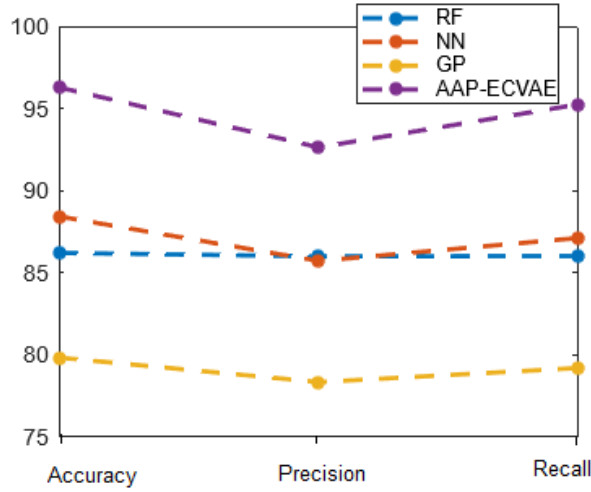


Figure 6: Overall performance metrics of Existing and suggested methods

Our main conclusions include a comparison of prediction interval lengths, Accuracy and coverage rate of the empirical prediction interval were calculated using the test hold-out set. As a measure of recall, accuracy, and precision, we compute the Advanced Archimedes principle coefficient of determination between the actual simulated sea-level change values and the calibrated ML model predictions.. In terms of prediction accuracy, Table 1 and Figure 6 clearly demonstrates that the recommended model AAP-ECVAE exhibits the highest accuracy, precision, and recall when compared to the RF, GP, and NN models.

5. Conclusion

In this paper the effectiveness of Machine Learning models, particularly the NN and AAP-ECVAE, in simulating variations in ocean level. Both the NN and the ECVAE offer precise prediction accuracy and good intervals of prediction. The AAP-CVAE and NN models' predictive performance is similar to a starting point. The accuracy of the GP model is comparable, but the basic GP model struggles to accurately measure uncertainty when it comes to coverage of interval for forecasting rates. We have also demonstrated the use of a simple Archimedes principle to infer statistically relevant uncertainty. We quantify this uncertainty using nonlinear predictions from machine learning. This method of calibrating uncertainty relies on the weather. In this study, the AAP-ECVAE methodology has produced accurate sea-level change prediction intervals simulations that are long enough to be practical, affordable to compute, and simple to use. According to this dissertation's research, earthquake activity in Japan and Antarctic ice melt are major events. The report's analysis evaluates Antarctic Ice Melt, Sea-Level Rise and Ocean Density Changes, Geological Stability, Antarctic Continental Shelf Fracturing, and Seismic Activity using the suggested model, demonstrating the highest prediction accuracy level. There are

difficulties associated with tsunami-induced seismic activity since secondary events might occur at any time and in any area. A thorough study of variables such as tsunami characteristics, fault stress states, and local geological structure is necessary to predict these occurrences. Future studies should concentrate on tectonic plate movements and stress patterns in relation to the Pacific Ring of Fire, the effects of climate change, and the melting of the Antarctic ice sheet. To meet these issues, weather experts, geologists, and politicians must collaborate across disciplinary boundaries.

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